

NAME: _____ CLASS: _____ INDEX: _____



CATHOLIC JUNIOR COLLEGE
JC2 PRELIMINARY EXAMINATION
Higher 2

ANSWERS

BIOLOGY

Paper 3 Long Structured and Free-response Questions

9744/03

23 AUGUST 2018

2 hours

Candidates answer on the Question Paper.

Additional Materials: Writing Paper

READ THESE INSTRUCTIONS FIRST

Write your index number and name in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs. Do not use staples, paper clips, glue or correction fluid. DO NOT WRITE IN ANY BARCODES.

Section A

Answer all questions in the spaces provided on the Question paper.

Section B

Answer one question in this section on writing papers provided.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together as follows:

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use	
1 [30]	
2 [10]	
3 [10]	
4 [25]	
5 [25]	
TOTAL P2 [35%]	75

This document consists of **22** printed pages and **0** blank pages.

[Turn over

Section A

Answer **all** questions in this section.

- 1 Foetal haemoglobin (HbF) is the main oxygen transport protein in foetuses during the last seven months of development in the uterus and persists in the newborn until six months old. Fig.1.1 shows a protein structure of HbF, which has a similar structure to HbA.

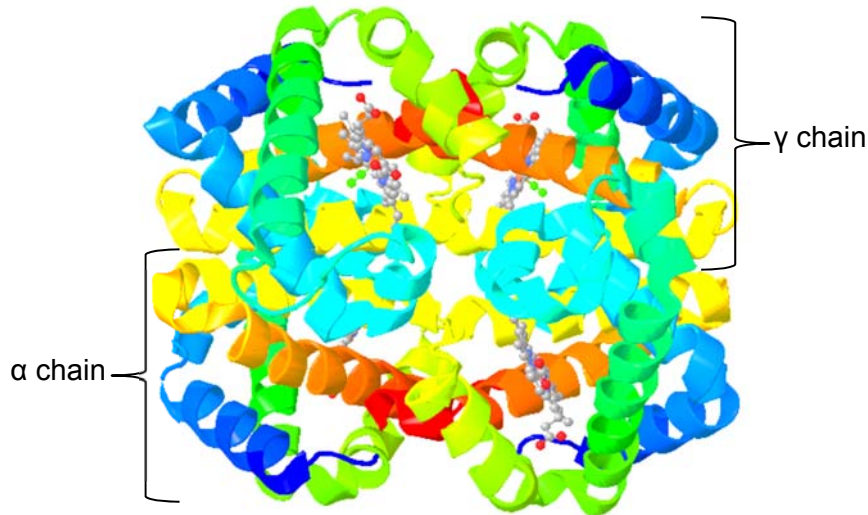


Fig. 1.1

(a)(i) With reference to Fig.1.1, describe the features within HbF that are similar to HbA.

ANS [L2] (Novel)

SC: with reference to Fig.1.1, describe
OR:

features of HbF similar adult form
Quaternary structure with 4 globin chains (2 alpha; 2 gamma) HbA
Centralised Haem group
variety of a.a. (hydrophobic , hydrophilic R groups) ● globular form

[2]

- Both HbA and HbF has **quaternary structure** with **4 globin chains** (i.e. **2α** chains and **2γ** chains for HbF)
- Both HbA and HbF has **a central haem group** in each of the globin chains for oxygen binding
- Both HbA and HbF are **globular** protein where the polypeptide chains are made up by variety of **amino acids with different R-groups** that folds into a **compact** shape exhibiting the various **interactions / bonds** (e.g. **H bonds / disulfide linkages / ionic bonds / hydrophobic interactions (at least name 2)**) to hold the protein structure in a **3D conformation shape / OWTTE**
- AVP**

Max 2

(a)(ii) HbF is a protein whose polypeptide chains are made up of amino acids that are joined together to achieve the final molecular configuration as shown in Fig. 1.1.

Using the monomer shown in Fig. 1.2, draw out a full labelled diagram to illustrate how the bond formation of bond is achieved between 2 such monomers.

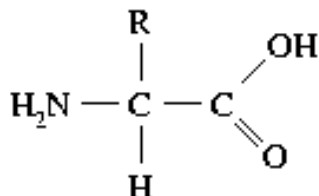
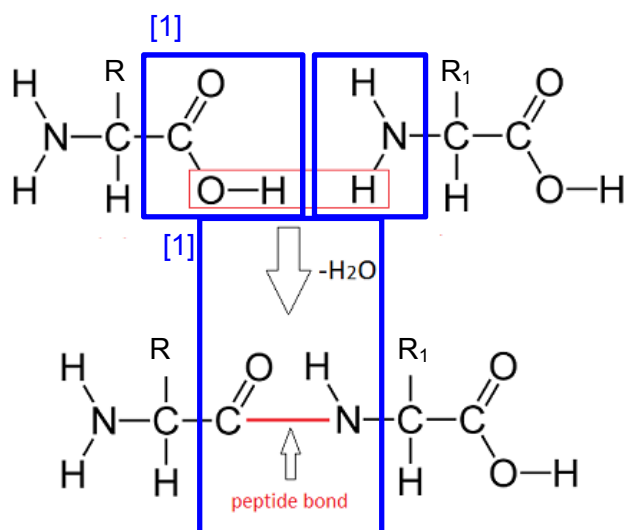


Fig. 1.2



[2]

ANS [L2] (Novel)

SC: Draw out labelled diagram formation of bond
OR:

[2]

[1] – condensation reaction with removal of water with a labelled peptide bond in dipeptide
(C=O and N-H)

[1] – correct structure of drawing, indicating –COOH and –NH₂ group as reacting groups

In newborns, foetal hemoglobin is completely replaced by adult hemoglobin (HbA) by approximately 6 months post-natal. Fig. 1.3 shows an oxygen saturation curve for HbF which appears left-shifted when compared to adult form (HbA). HbF is better able to bind to oxygen compared to HbA, giving better access to oxygen from the mother's bloodstream.

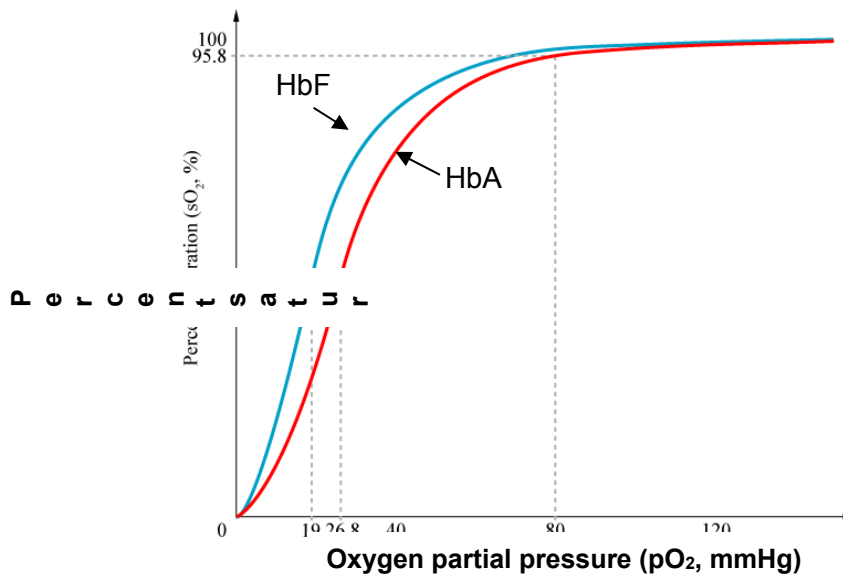


Fig.1.3

(a)(iii) With reference to Fig.1.3, suggest the significance of HbF possessing a higher capacity in binding to oxygen compared to HbA.

ANS [L3] (novel)

SC: With reference to Fig. 1.3, suggest
OR:

significance HbF higher capacity
[O₂] high outside compared to bloodstream
higher affinity in O₂ for HbF

[2]

1. Oxygen concentration from the environment is **higher** compared to oxygen concentration in the mother's bloodstream.
2. Foetus needs to have a better access in taking in the oxygen, which is already at a lower concentration, and hence requires a **higher affinity** for the HbF to bind / absorb oxygen, compared to HbA
3. **AVP**

Formation of globin chains varies in foetal to adult life. The alpha (α) and beta (β) and to a lesser extent the delta (δ) globin chains form HbA. In foetus, the gamma (γ), epsilon (ϵ), and zeta (ζ) chains are present. Fig. 1.4 shows the β -like globin genes that are expressed in a developmentally restricted manner, which is located on chromosome 11, as depicted in locus control region as well as the relative proportions of the various globin chains from conception to six months of age.

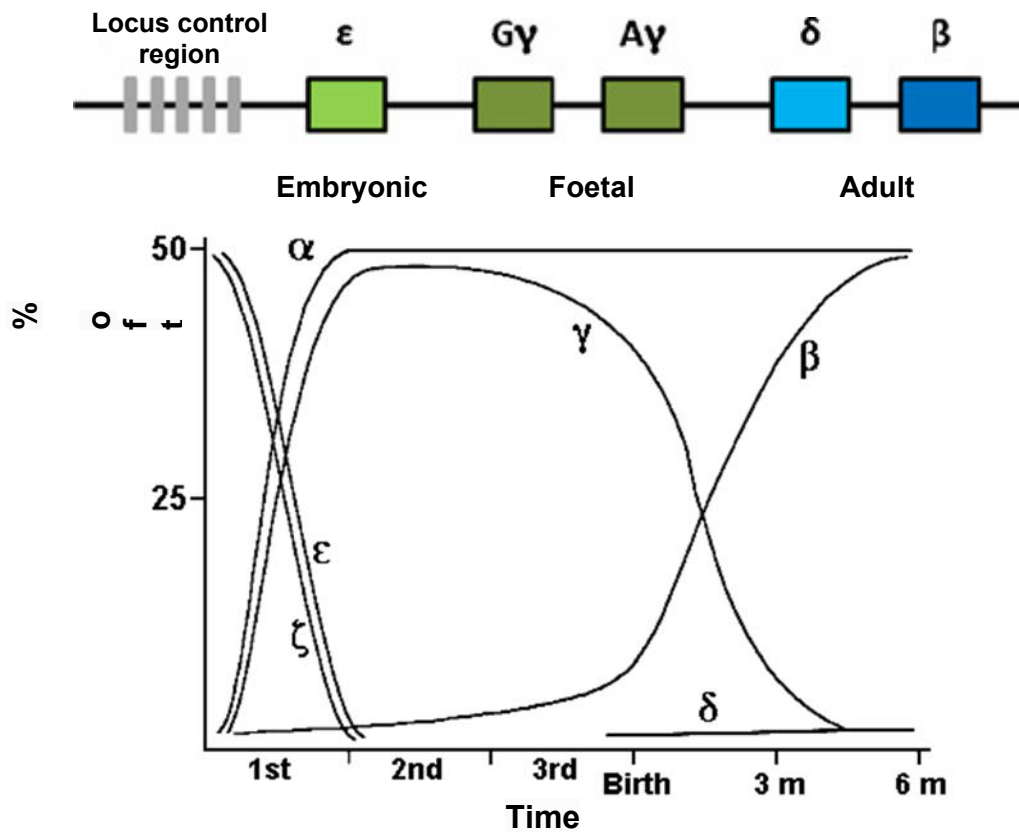


Fig. 1.4

(a)(iv) With reference to Fig.1.4, explain how the globin genes are regulated at transcriptional level to produce HbF and then HbA.

ANS [L3] (Novel)

SC: With reference of Fig.1.4 explain how OR:

globin genes regulated @transcriptional level

1. gene silencing • prevent expression of certain genes @ different developmental stages
2. Transcriptional activator proteins and transcriptional repressor proteins + enhancers & silencer sequences • to regulate globin gene expression
3. switch on α and γ genes • HbF at foetal stage up until birth
4. switch on α and β genes • HbA after birth

[2]

Globin genes are regulated through **gene silencing** / prevention of expressing certain genes at different developmental stages / OWTTE

1. 2.1 This is by switching on α and γ globin genes to form HbF protein molecule at foetal stage up until birth
OR
2.2 switching of α and β globin genes to form HbA protein molecules after birth onwards
(Accept reverse argument)
2. 3.1 **Transcriptional activator proteins** and **transcriptional repressor proteins** (accept **Specific transcription factors**) binds to respective **enhancer** and **silencer** sequences to activate / repressed α and γ globin genes to form HbF
OR
3.2 **Transcriptional activator proteins** and **transcriptional repressor proteins** (accept **Specific transcription factors**) binds to respective **enhancer** and **silencer** sequences to activate / repressed α and β globin genes to form HbA

Sickle cell anaemia is an inherited disease where the red blood cells are most commonly caused by the haemoglobin variant HbS, where a mutation has occurred at the beta globin polypeptide chain. Figure 1.5 shows the blast sequence of both the normal adult haemoglobin variant, HbA, as well as the mutant adult haemoglobin variant, HbS.

Beta globin sequence in normal adult haemoglobin (HBA)

nucleotide base	CTG ACT CCT GAG GAG AAG TCT
amino acid	Leu – Thr – Pro – Glu – Glu – Lys – Ser

Beta globin sequence in mutant adult haemoglobin (HBS)

nucleotide base	CTG ACT CCT GTG GAG AAG TCT
amino acid	Leu – Thr – Pro – Val – Glu – Lys – Ser

Fig. 1.5

(b)(i) With reference to Fig. 1.5, suggest how the mutation could have occurred.

ANS [L2] (adapted from CA3)

[2]

SC: with reference to Fig.1.3, suggest how
OR: cause?

mutation could have occurred

1. error in DNA replication ● lack of proofreading by DNA pol I and III
2. replication errors ● not detected
2. mutations in genes DNA repair enzymes ● not detected & no repair function

3. During DNA replication process, **DNA polymerases III** are responsible for the **elongation** of the strand makes **mistakes** in the **base pairing**.
4. DNA polymerase I which has the **proofreading function**, **fails to detect** the error and correct it

OR

5. Due to **replication errors** / wobble-induced replication errors / strand slippage replication errors often happens and hence, any mismatched base pair are **not detected** and **not repaired**
6. If not corrected, the affected DNA strands will become **templates** in the **next round of replication** and the **error is** passed on to daughter cells / **OWTTE**

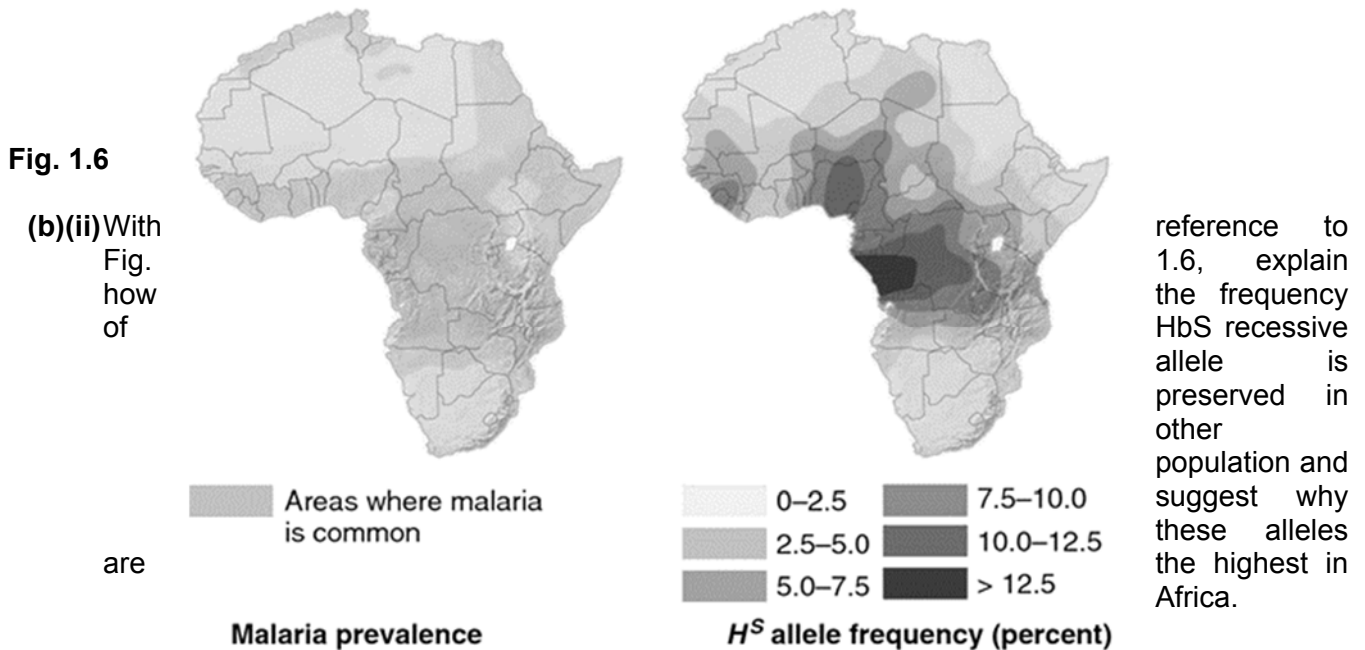
OR

7. **Mutation** can occur in the **genes** that **codes for DNA repair enzymes** causing these enzymes to be **non-functional** and **cannot** perform their **repair** function and hence, any mismatched base pair are **not detected and repaired**
8. If not corrected, the affected DNA strands will become **templates** in the **next round of replication** and the **error is** passed on to daughter cells / **OWTTE**

Reject UV light / mutagens

Sickle cell disease has its highest occurrence in Sub-Saharan Africa. Despite the survival disadvantage of the allele HbS, HbS allele persists in the population at more than 12.5%.

Fig. 1.6 shows regions where sickle cell anaemia is most common, which coincide with regions where malaria is. Malaria-causing parasites are introduced into the blood by mosquitos.



ANS

(Novel)

SC: With reference to Fig.1.6

1. explain how
2. suggest why

HbS frequency
these alleles highest

preserved

[L2/L3]

[4]

OR:

1. Explain why
 - Diploidy ● HbS: recessive allele ● exist in carriers (Hb^AHb^S) and passed down thru generations
 - hence maintain pool of HbS in the population
2. Heterozygous protection
3. Suggest why
 - selection pressure (malaria) is the greatest; heterozygous individuals (selective adv)
 - need to survive malaria ● HbS allele is selected for (for malaria parasite can't develop in sickle cells ● X infection)
 - need to survive from SCA ● HbA allele is selected for (for efficient O₂ transport at low [O₂] levels)

[Explain why]

1. Due to diploidy, recessive alleles such as HbS are **hidden / masked over** by the **dominant allele** HbA. and are thus allow the individuals to survive and the allele persist.
2. **Heterozygous protection**
3. Thus, it always existed **in carriers** (Hb^AHb^S) and passed down through generations, **maintaining the gene pool of HbS in the population**

(Max 2)

[Suggest why]

4. **Selection pressure** to survive from **malaria** is greatest (shown in Fig.1.6), thus **heterozygous** individuals are at **selectively advantage**.
5. There is a need to survive from malaria where **HbS is selected** for as **malaria parasite cannot developed in sickle cells**.
6. There is also a need to survive from sickle cell anaemia (SCA) where **HbA is selected** to allow **efficient oxygen transport** around the body, especially **at low oxygen concentrations**.

(Max 2 marks for points 4 - 6)

In adults, foetal hemoglobin production can be reactivated and induced pharmacologically, which is useful in the treatment of diseases such as sickle-cell disease. When present in individuals with sickle cell anaemia, HbF prevents sickling. Besides induced HbF treatment, haematopoietic stem cell transplantation (HSCT), usually derived from bone marrow, peripheral blood or umbilical cord

blood has also shown great progress in the treatment of sickle cell disease recently. This involves in either the extraction of stem cells from various sources which may be autologous (i.e. using the patient's own stem cells are used), allogeneic (i.e. using stem cells derived from a donor) or syngeneic (i.e. using stem cells from an identical twin). Researchers cultivate these extracted stem cells which is then injected back into the patient. Such therapy for sickle cell anaemia could potentially give new hope to patients with sickle cell anaemia, although the treatment comes with its own risk.

(c)(i) Describe the unique features of stem cells used to treat this disease.

ANS [L1] (Novel)

[2]

SC: describe unique features of SC treat THIS disease
OR: unspecialised, undifferentiated, proliferate continually, self-renewal

1. They are **unspecialized cells** without any tissue-specific structures and
2. are **multipotent**, hence they are able to be induced to **differentiate** into **blood cells** ;

(c)(ii) Explain why HSCT treatment from allogeneic sources may pose some risks to the patient.

ANS [L2] (Novel)

[2]

SC: Explain why HSCT treatment pose risk
OR: Tissue rejection by recipient patient
Graft-vs-Host disease ● WBC from donor cells 'attack' recipients' host cells (new info.)

Either

1. Recipient host cells considers **donor stem cells as foreign** and
2. **rejects** by host's immune system, which then **destroy the transplanted tissue/cells**. / OWTTE

Or (more accurately)

1. White blood cells from donor's immune system remain within donated cells/tissues cells **recognise the recipient as foreign** and
2. **attack** the recipient's body cells, which leads to graft-vs-host disease / OWTTE

(c)(iii) Suggest why extraction of haematopoietic stem cells from autologous sources may not be as effective for therapy.

ANS [L3] (Novel)

[1]

SC: Suggest why extraction of stem cells autologous sources not effective
OR: SC from own patient
1. SC not effective ● for dominant allele type of diseases
2. inherited a defective gene ● cannot overcome the problem

1. The genes in stem cells from autologous sources are genetically similar and flawed therefore it is ineffective as a therapy for HbS. / OWTTE

(c)(iv) Umbilical cord blood has a higher concentration of haematopoietic stem cell than normally found in adult blood. Just in 2014, Singapore Cord Blood Bank (SCBB) received a 67% increase in cord blood donation compared to the year 2010. Nowadays, many expectant mothers and couples opt in banking their child's cord blood for many reasons.

Suggest the advantages of banking baby's cord blood.

ANS [L3] (novel)

[3]

SC: Suggest advantages banking baby's cord blood
OR: therapeutic application
transplant histocompatibility
Ready source / availability of SC

1. Umbilical cord contains stem cells which are **pluripotent / multipotent** which can **differentiate into several cell types**
(max 1)
2. Stem cells from cord blood can be used to **treat patients** who have **undergone chemotherapy** for cancer that **destroy their bone marrow** or other blood-related **disorders**
3. Stem cells can undergo **transdifferentiation**, under controlled conditions to become other cell types for **regenerative medicine / cell replacement therapy**
(max 1 – for either points 2 or 3)
4. Stem cells are **genetically identical** and will **not trigger immune response** when introduced into the **same** person in future
5. Such stem cells **increases chances of a match or histocompatibility** should his/her siblings/parents needed **stem cell therapy**
(max 1 – for either points 4 or 5)
6. A **safe, painless** and **non-controversial** means of acquiring stem cells.
7. Readily available should need arise, **eliminating time needed to find a matching donor's stem cells** and **allowing treatment immediately / OWTTE**
(max 1 – for either points 6 or 7)

MAX 3

The signals governing self-renewal versus multi-lineage differentiation of hematopoietic stem cells (HSC) are complex. Several key signaling pathways including BMP, Hedgehog, Notch, Wnt, TGF-beta and others play important but different roles during foetal hematopoietic development, normal bone marrow haematopoiesis (i.e. formation of blood cellular components) and leukemogenesis (i.e. development of leukaemia). Haematopoiesis is controlled by combined effects of growth factors that permit cellular proliferation and transcription factors that activate lineage-specific genes. mRNA transcripts resulted from the these transcription of cell-specific genes must be successfully processed before they can be exported out of the nucleus, as shown in Fig. 1.7.

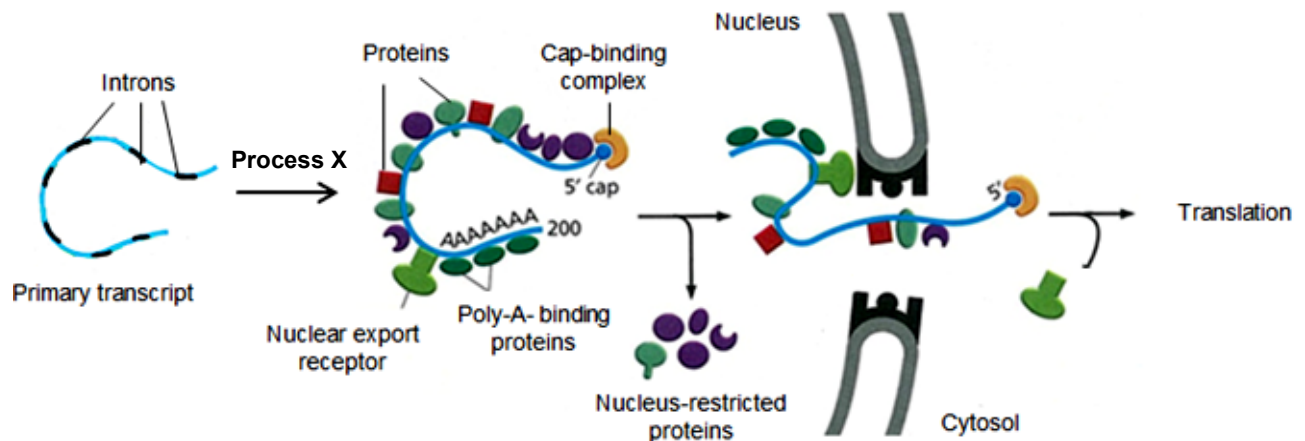


Fig. 1.7

With reference to Fig. 1.7,

(d)(i) Describe process X that results in the formation of a continuous mRNA coding sequence

ANS [L2] (modified H2 HCl/P2/Q2)
 SC: describe process stage X
 OR:

premRNA → mRNA
 RNA splicing by spliceosome
 introns excised; exons spliced tgt

[2]

1. **RNA splicing** occurs in process X, which is carried out by **spliceosomes**
2. where **introns are excised** and **exons are spliced together** to form a contiguous **mature mRNA**.

(d)(ii) Explain the significance of process X in regulating translation of mRNA.

ANS [L2] (novel)

SC: Explain significance process X control @ translation stage
 OR:
 1. Cap binding complex • small subunit of ribosome can recognise 5'end of mRNA
 2. Cap binding complex & poly A binding proteins • ensure mRNA half-life in nucleus, protect against degradation by nucleases

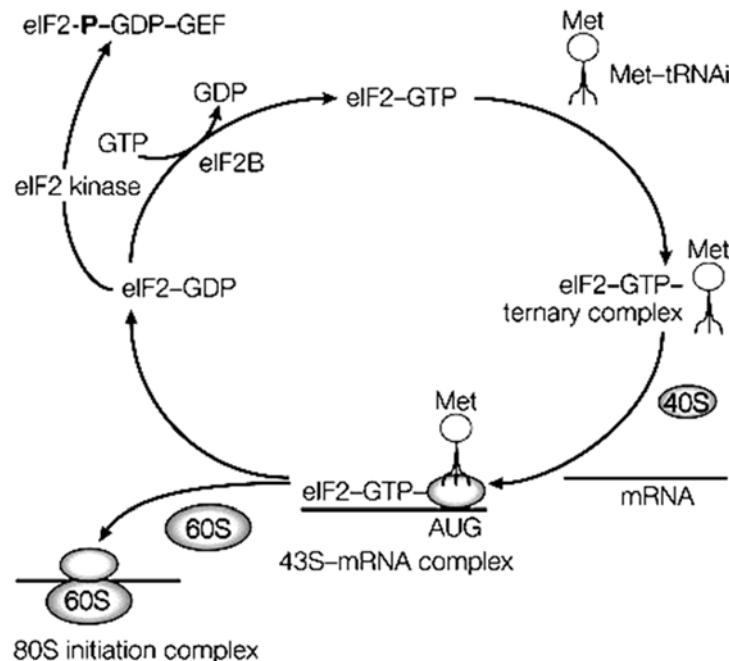
[2]

1. **Small subunit of ribosome** can **recognise the 5'end of mRNA** through the **Cap binding protein**.
2. It is to ensure **mRNA half-life in nucleus** where there is a need to **protect mRNA from degradation by the nucleases**. / OWTTE

Translational control plays a crucial role in gene expression regulation, especially defining the proteome, maintaining homeostasis and controlling cell proliferation, growth and development. Translational control governs the efficiency of mRNAs. There are several ways in how mRNA translation can be regulated. One of the best studied examples include phosphorylation of eIF2 α (a subunit within eIF2 protein), eukaryotic initiation factor 2, which results in an inability for tRNA-binding methionine (met-tRNA) from binding to ribosomes. eIF2 is an essential factor for protein synthesis that forms an initial complex with initiator Met-tRNA, using GTP as energy source. This is shown in

Fig. 1.8.

Fig. 1.8



(d)(iii) With reference to Fig. 1.8, suggest how phosphorylation of eIF2 α of eIF2 protein may affect initiation of translation.

ANS [L3] (Novel)

SC: With reference to Fig.1.8

OR:

suggest how phosphorylation of eIF2 affect initiation translation
 1. eIF2 α -P cannot recognise initiator Met-tRNA; initial complex cant form
 2. Cannot bind well on mRNA template near 5' cap ● no translation initiation complex
 ●no translation

[2]

1. eIF2 α phosphorylation cause eIF2 protein **cannot recognise initiator Met-tRNA**, hence **initial complex cannot be formed**
2. This leads to complex **not able to bind well on mRNA template** near 5' cap, **disrupting / preventing** formation of **translation initiation complex** and hence no translation took place
OWTTE

(d)(iv) Suggest one other method that can be used to control translation and explain how it achieves the outcome.

ANS [L3] (Novel)

SC: Suggest

OR:

one other outcome
using microRNA(miRNA)

control translation

1. destabilising mRNA via shortening of polyA tail
2. mRNA translation hindered by ribosomes

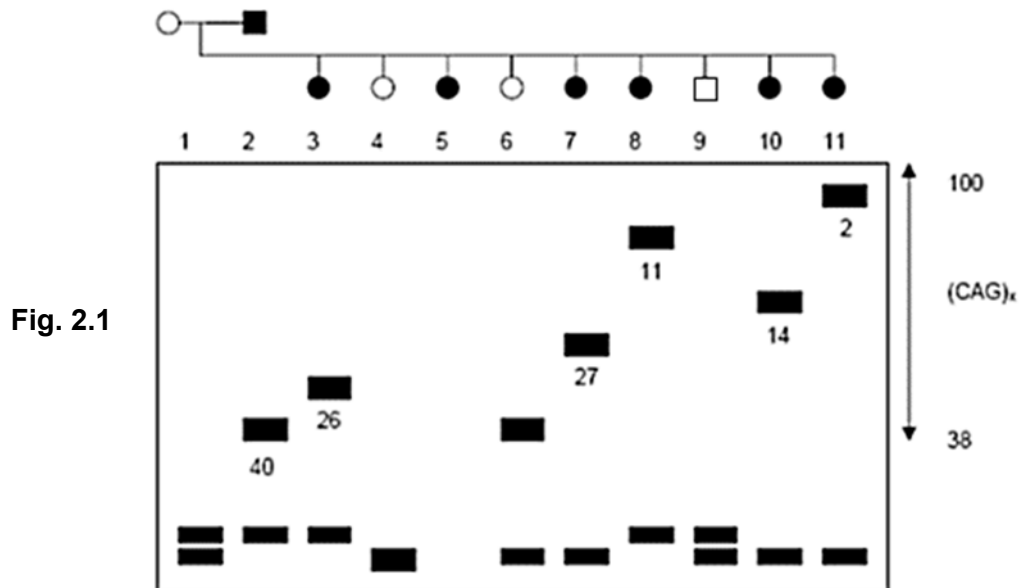
[2]

3. [Method] Use **microRNA (miRNA) / interference RNA (RNAi) / RNA oligonucleotides** that can undergo complementary base pairing with complementary mRNA
4. [Outcome]
 - Destabilising mRNA strand via shortening of polyA tail
 - mRNA translation is hindered by ribosomes
 - **AVP**

[Total: 30]

- 2 Huntington's disease is an inherited disease that causes the progressive breakdown (degeneration) of nerve cells in the brain. Huntington's disease has a broad impact on a person's functional abilities and usually results in movement, thinking (cognitive) and psychiatric disorders.

The data below shows the results of electrophoresis of PCR fragments amplified using primers for the site which has been shown to be altered in Huntington's disease. The inherited mutation in the Huntington's disease gene on chromosome 4, abnormally repeats the nucleotide sequence CAG from 36 to more than 120 times. The male parent, as shown by the black box, got Huntington's disease when he was 40 years old. Six of his children (individuals 3, 5, 7, 8, 10, 11) suffer from Huntington's disease, and the age at which the symptoms first began is shown by the number below the band from the PCR fragment.



(a) to Fig. 2.1, explain how Huntington's disease is inherited.

With reference

ANS [L2] (Novel)

SC: ref 2.1

OR:

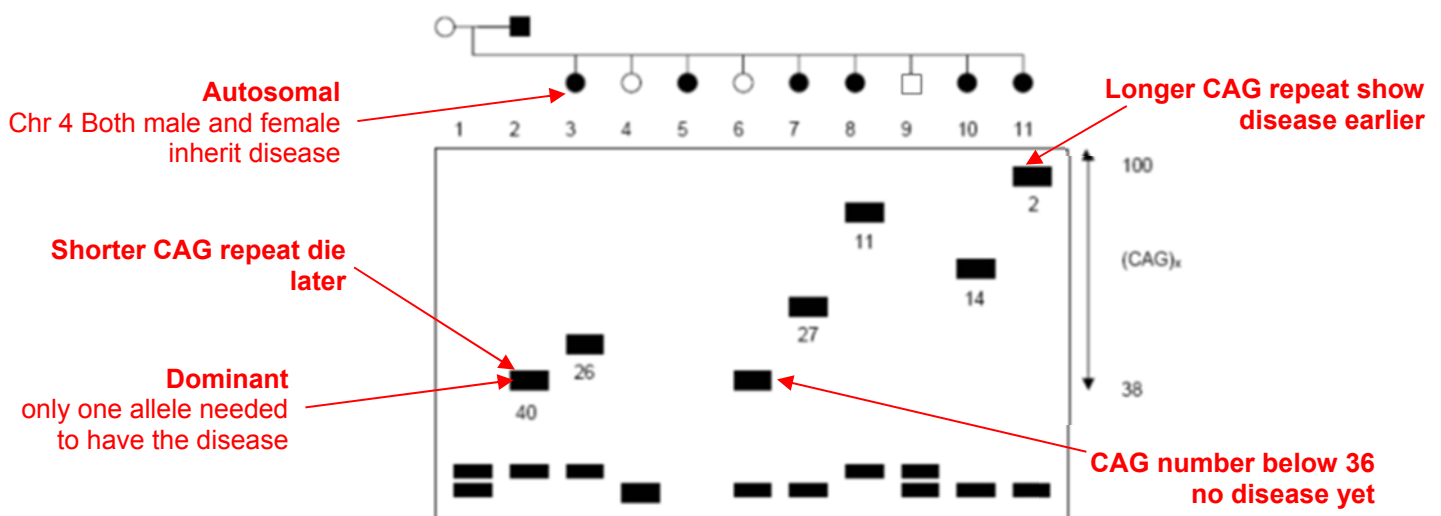
how Huntington's disease

1] Autosomal – both male and females with disease.

2] Dominant- only one allele needed to develop disease. 2/3/5/7/8/10/11

inherited

[2]



1. **Autosomal – on chromosome 4** both males and females inherit the disease and Or it is autosomal because- if it was sex linked dominant ALL MALES would be normal and ALL FEMALES diseased which is not the case.
2. **Dominant** only one allele need be present indiv 2/3/5/7/8/10/11.

OWTTE. Marks given even w/o citing reference.

(b) (i) What is the genotype of individual 4, explain why.

ANS [L2] (Novel)

[2]

SC: ref 2.1 genotype indiv 4 explain
OR: 1] homozygous recessive 2] 2 copies of smaller fragment denoted by thicker band

1. **Homozygous recessive**
2. copies of **smaller fragment denoted by the thicker band.**
3. No presence of >36 CAG long strand.

Max 2

(b)(ii) With reference to Fig. 2.1 describe a correlation between PCR fragment length and Huntington's disease. Explain your answer.

ANS [L2] (Novel)

[2]

SC: ref 2.1 correlation PCR fragment Huntington's disease
OR: 1] Longer the length earlier onset of disease
2] indiv 11 100xCAG long disease at 2
3] indiv 2 38xCAG long disease at 40

1. The longer the PCR fragment length above 36 CAG- the earlier the onset of the disease
2. **indv 11** 100xCAG long disease at age 2 **and** **indv 2** 38xCAG long disease at age 40.

Show trend OWTTE

(c) With reference to Fig. 2.1 explain the abnormality in individual 6.

ANS [L3] (Novel)

[2]

SC: ref 2.1 abnormality in indiv 6
OR: 1] heterozygous- diseased phenotypically normal
2] disease onset not occurred yet, CAG fragment under 37

1. Indv 6 diseased but is phenotypically normal OWTTE
- OR
2. indiv 6- has **not shown the onset of the disease**
- and
3. **CAG length below 37 times**

(d) Explain one further test on the product of Fig. 2.1 that can be employed to confirm the genotype of individual 6.

ANS [L3] (Novel)

[2]

SC: ref 2.1 further test confirm genotype indiv 6
OR: 1] Nucleic acid hybridization / probe specific sequence of disease complementary base paired.

1. Nucleic acid hybridization.
2. indiv 6- **CAG length below 36 times**

[Total: 10]

- 3 Humans are hypothesized to share a common ancestor with modern great apes; however, humans have 23 pairs of chromosomes while great apes have 24 pairs. In 2005, one paper was published regarding the analysis of the genomes of humans and chimpanzees where they were compared for homology (Fig. 3.1). An evidence that strongly supports the shared ancestry is chromosome number 2 in humans, which was found to be the result of an end-to-end fusion of two ancestral chromosomes that remain separate in other great apes.

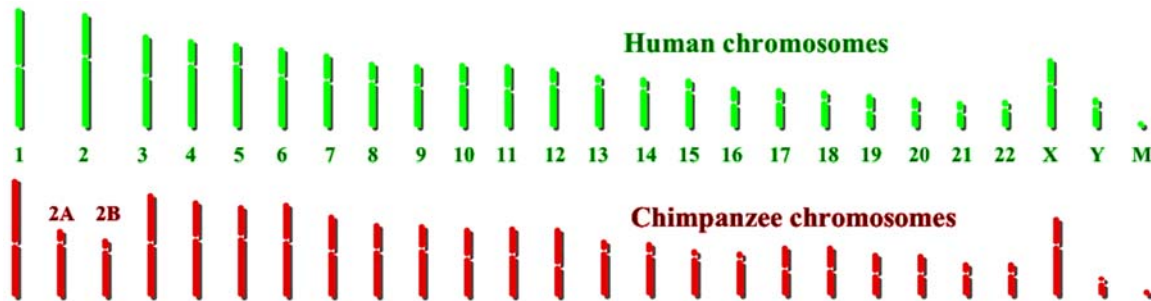


Fig. 3.1: Comparison of chromosomes 1 to 23 of humans and 1 to 24 of chimpanzees, M representing mitochondria.

- (a) Outline how fusion of chromosomes could possibly occur.

ANS [L2] (H2 Novel)

[3]

SC: Describe how fusion processes chromosomal mutation possibly occur any causes for (not just limited to chromosome 2)

OR: 1) mutation resulting in chromosomal aberration
2) Translocation
3) Crossing over errors

1. Fusion of chromosomes can be a result of **chromosomal aberration** where a kind of **mutation** has occurred;
2. **Translocation** has occurred where **one chromosome was cut and joined to another chromosome**;
3. **Addition of chromosomes** pieces / segments to the same homologous chromosome as a result of **crossing over** and chromosomes segments remaining joined
4. **Wrong pairing of chromosomes during meiosis I** and resulting in **crossing over** and chromosomes remaining joined

- (b) Scientists hypothesised that humans and great apes are highly likely to share a common ancestry. By comparing the genome of humans against great apes, especially regarding the structure of chromosome number 2 in humans, they were able to find more evidences that the hypothesis should be true.

Predict what you would expect to find from this study if the hypothesis is true.

ANS [L3] (H2 Novel)

[4]

SC: based on info predict expected if hypothesis true
 Chromosome 2 joined end to end similar homology between humans and apes
 OR: 1) length of joined chromosome roughly similar to separate
 2) gene sequences very similar
 3) 2 centromeres
 4) joined telomeres

1. **Total length of the two separate chromosomes** in primates would be similar to the **length of chromosome 2**;
2. The **gene sequence of chromosome 2** should be **very similar to the gene sequences of two separate chromosomes** in primates;
3. **Chromosome 2** would also contain **an extra centromere** in its structure;
4. **Chromosome 2 also should also have telomere sequences in the middle** in addition to the both ends;
5. Primates should share **all similar chromosomes as humans, except for one additional** (pair of) **chromosome** that is found in chromosome 2 of humans

- (c) Some animals such as mice and non-human primates (NHPs) are used in clinical research because of their relatedness to humans as seen in Fig. 3.2. However, there are strong concerns and strict regulations regarding the ethical use of NHPs in research, especially when compared to other animals. NHPs are used in clinical research only when deemed necessary, such as during the *Ebola* outbreak when they were used for further testing for vaccines after testing the same vaccines on mice.

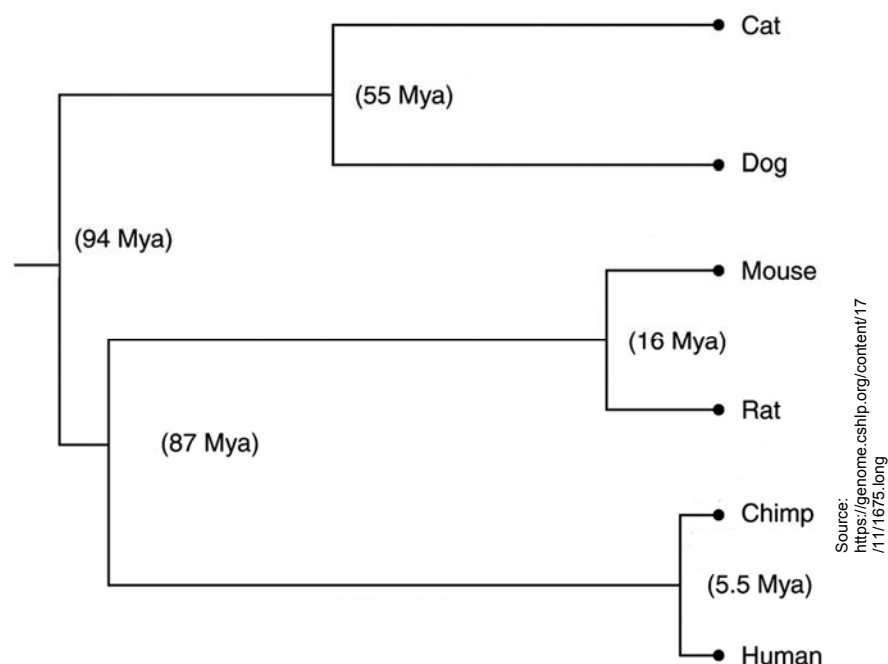


Fig. 3.2: Phylogenetic tree showing relative closeness of different mammals, where Mya = millions of years ago where lineage diverged.

- (i) Suggest why it is necessary to use non-human primates (NHPs) for further clinical research after testing on animals such as mice.

ANS [L3] (H2 Novel)

[1]

SC: Suggest why necessary to use NHPs
OR: Give reasons ensure safety as mice not close enough / NHPs are closer

1. To ensure safety of the vaccines as NHP cells are much more similar to human cells and therefore would provide results / respond to treatments more accurately / more similarly to humans as compared to mice
2. Accept reverse argument: Results from mice may not be similar enough to responses from human cells as compared to NHPs

(ii) Discuss why the use of NHPs in scientific research raises such strong ethical concerns.

ANS [L3] (H2 Novel)

[2]

SC: discuss why (reasons) NHPs raises strong ethical concerns
OR: high similarity between humans and primates
1] high intelligence
2] display high level of emotions
3] unable to give consent

[Any 2]

1. Due to the high homology between humans and primates, primates are considered to have high intelligence;
2. hence they display a high level of emotions.
3. they are likely to suffer in similar ways as humans / OWTTE
4. They have the capacity to make choices and decisions, but are unable to consent to research
5. AVP

Section B

Answer **one** question in this section

Write your answers on separate answer paper provided.
Answer each part on a **separate** piece of paper.

Your answer should be illustrated by large, clearly labelled diagrams, where appropriate.
Your answer must be in continuous prose, where appropriate.

Your answer must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 4 The actions of humans have had numerous impacts on both the environment and the natural world, affecting both plants and animals.

- (a)** One such impact is climate change which affects plants in many ways, including photosynthesis.

Outline the main stages of photosynthesis and suggest how anthropogenic climate change may potentially affect this process in plants. [14]

ANS [L2/L3] (H2 Novel)

SC: Outline main stages of photosynthesis
ONLY briefly describe important steps

- OR:
- 1] Light-dependent
 - a) Absorb light, excite e-
 - b) Pass down ETC, pump H⁺
 - c) Produce ATP & NADPH
 - 2] Light-independent
 - a) Carbon fixation
 - b) Production of glucose

[14]
suggest how CC affect photosynthesis
cause – effect on components / processes in photosynthesis

- 1] Effects of CC: CO₂ levels, temperature, water
- 2] Overall increase in substrate CO₂
- 3] Temp affects ETC proteins,
- 4] ETC membrane,
- 5] Calvin cycle enzymes,
- 6] Water affects transpiration and stomata
- 7] Different plants affected differently

1. In the **light-dependent reactions**, **energy from sunlight is absorbed by chlorophyll** and an **electron** in chlorophyll a molecules (P680 / P700) is **excited to a higher energy level**.
2. **Electrons** travel down the chain from **one electron carrier to another electron carrier** in a series of redox reactions **of decreasing energy**, losing a small amount of energy each time they move from one electron carrier to another.
3. As electrons pass through the electron carriers, **H⁺ are pumped into the thylakoid lumen**, contributing to the **proton gradient** and thus **proton motive force** that is subsequently used in **chemiosmosis**.
4. The energy **released** from the **electron transfer** along the **ETC** is used to **synthesise ATP** from ADP and Pi (inorganic phosphate).
5. Fd then passes the electron to **NADP**, acting as a **final electron acceptor, producing NADPH (reduced NADP)** on the side of the membrane facing the stroma.
6. The **Light-Independent Reactions**, also known as the **Calvin Cycle**, occurs in the stroma of the chloroplast, utilizing the **ATP** and **NADPH** synthesized **from the light-dependent reactions**.
7. **Carbon fixation** is catalysed by **enzyme Rubisco** where Ribulose biphosphate (RuBP) reacts with carbon dioxide (CO₂) to form 3-phosphoglycerate (PG/PGA)
8. **PG is phosphorylated by ATP** into 1,3-bisphosphoglycerate (BPG) which is reduced by **NADPH to triose phosphate** which are **eventually converted to glucose** and then sucrose, which are then transported in the phloem from leaves

[Max 6]

Effects of climate change:

9. Due to human actions such as burning of fossil fuels, there is an **increased level of carbon dioxide in the atmosphere** which also resulted in an **increase in temperature** as a consequence.
10. Due to the increase in temperature / global warming, there is an **expected decrease in frequency of rainfall** resulting in potential droughts and an **increase in volume of precipitation / rainfall** resulting in potential floods.

Effects on photosynthesis:

11. **Increased levels of carbon dioxide will directly increase the rate of carbon fixation in photosynthesis** as the concentration of substrate available for the enzyme increases;
12. There is an **increase rate of enzyme-substrate complex formation** between Rubisco, RuBP and CO₂
13. Generally, **rate of photosynthesis will increase** as **temperature increases slightly** as multiple enzymes are involved in the process of photosynthesis and there will be higher frequency of successful collision between Rubisco, RuBP and CO₂ / cite relevant reaction

Reject: **increased light absorption**

14. However, as **temperature increases** faster than plants can adapt / increases more drastically, **enzymes** involved in photosynthesis are more likely to **denature** and hence **rate of photosynthesis will decrease**.
15. Increase in temperature will also **increase the fluidity of the thylakoid membrane** which may **increase leakiness of the H⁺ protons** resulting in overall less chemiosmosis and less ATP produced (Sharkey, 2005, <https://onlinelibrary.wiley.com/doi/pdf/10.1111/j.1365-3040.2005.01324.x>)
16. Increased fluidity of thylakoid membrane may also **result in inefficient electron transference of energy** due to the **proteins not being in close proximity**
17. Decrease in rainfall and increase in precipitation will **result in overall less availability of water to plants**
18. Plants will need to compensate by **reducing stomata density / time when stomata are open to absorb CO₂** and this **may limit the amount of carbon dioxide intake** despite the increase in concentration of CO₂
19. Thus, rate of photosynthesis is **likely to increase for plants found in colder climate** as temperature is less likely to increase to a point where enzymes denature but rate of photosynthesis is **likely to decrease for plants already found in warmer climates**.
20. Increased temperature will affect temperate plant and tropical plants differently; temperate plants may be able to photosynthesise more but tropical plants are likely to experience lower rate of photosynthesis

[Max 9]

QWC [1]; At least main stages of photosynthesis that climate change will affect are cited in first part of answer + at least 1 point on how climate change will affect photosynthesis.

- (b) Many modern domesticated animals are the result of artificial selection and breeding by humans over hundreds of years from wild animals. This includes chickens which were selected from jungle fowls for food and dogs which were selected from wolves to be kept as pets. Many of these domesticated animals look very different from their wild counterparts.

Explain how artificial selection has resulted in domesticated animals such as those mentioned above and comment on the view that “domesticated animals should be considered different species from their wild counterparts”. [11]

ANS [L3] (H2 Novel)

[11]

SC:	Explain how artificial selection resulted Focus on process of evolution	comment on view For / against	considered different species using different species concept
OR:	1] Genetic variation 2] Artificial selection (replacing natural selection) 3] Like produce like 4] Over time, allele freq changes	1] Morphological: different species 2] Biological: depends a) IF can breed, same b) IF cannot breed, different 3] Genetic: depends on degree of differences; rather arbitrary 4] Phylogenetic: same species	

- From the **original population** of wild animals, there was **genetic variation** within the population.
- Particular traits were selected for by humans**, e.g. chicken: plump flesh, egg-laying, dog: appearances, obedience, ability to guard, which **acted as selection pressure** on the population of wild animals.
- These **selected animals were then bred** (accept: reproduce) to **produce more offspring** that had **similar traits** selected for
- Animals that **did not carry the desired traits were selected against** and were **not allowed to breed**.
- Over several generations**, the same traits were selected for by humans, resulting in an **increase in allele frequency / mutations** that expressed these traits.
- Hence **modern domesticated animals** that were **selectively bred for** now mostly **carry the alleles and / or mutated genes** that **express the selected traits** which are very different from their wild counterparts.
- The **selection pressure in the wild** is very **different** from the artificial **selection pressure** on the domesticated animals and **hence these domesticated animals look very different from their wild counterparts**.

[Max 5]

- Whether these domesticated animals should be considered different species from their wild counterparts is **difficult to judge**, as the decision is **dependent on the species concept being utilised to classify them**;
- Under the **morphological species concept**, because these **domesticated animals look very different from their wild counterpart**, they should be **considered different species**;
- Under the **biological species concept**, because these domesticated animals **do not encounter their wild counterparts anymore / domesticated and wild counterparts may not recognise each other anymore**, they face **reproductive isolation** and hence should be **considered different species**;
- Under the **biological species concept**, if these animals can **still interbreed with their wild counterparts / have the potential to breed with their wild counterparts**, then they **should not be considered different species**;

12. Under the **genetic species concept**, if these animals **are genetically drastically different** from their wild counterparts due to possible accumulation of mutations, they should be **considered different species**;
 13. Under the **genetic species concept**, while it is clear to use molecular data to decide whether they are the same species, the **decision can be quite arbitrary** because scientists are not agreed upon the **percentage differences the domesticated animals should have from their wild counterpart** to be considered as different species (species problem);
 14. Under the **genetic species concept**, if **animals are still genetically very similar to their wild counterparts** but only have different alleles, they should **still be considered similar species**;
- [10-12 are all accepted being “if” scenarios describing different domesticated animals e.g. chickens are considered same species as jungle fowls but wolves and dogs are considered different species]**
15. Under the **phylogenetic species concept**, because **these domesticated animals all share a common ancestor with their wild counterparts**, although they may look quite different, they should be considered the same species;
 16. The view is even more difficult to support as **different domesticated animals can be considered same or different species** e.g. chickens are still same species while dogs are different species even if utilising the same species concept; domesticated animals should not be lumped together;

[Max 6]

QWC [1m]; process of artificial selection explained and at least 1 sound species concept used to consider speciation

[Total: 25]

- 5 (a) Compare the advantages and disadvantages of the different reproductive cycles of lambda phage and influenza. [13]

ANS [L3] (Novel)

SC: Adv / Disadv Life cycles Lambda phage Influenza
OR: Compare on the same level.

[13]

Approach	Lambda phage	Influenza
Mode of recognition of host cell	A1] Advantage in Specificity in binding to host. Specific protein receptors on tail fibers of lambda phage allowing virus to recognize specific host cells and bind securely.	A2] Advantage in Specificity in binding to host. Specific protein receptors / glycoprotein on viral envelope membrane - hemagglutinin binds to sialic acid of host membrane. allowing virus to recognize specific host cells and bind securely.
Mode of entry / penetration	D3] Disadvantage in Forced entry into host / bacteria - mode of entry may initiate host damage , D4] After the tail pins on baseplate is firmly on the cell surface, conformation changes occur in the baseplate and sheath.- Sheath contracts and tail tube penetrates outer membrane. D3] A phage enzyme (lysozyme) "drills" a hole in the bacterial wall. Pilot protein helps phage DNA to cross inner membrane.	A5] Advantage as specific binding of hemagglutinin to sialic acid does not affect the integrity of host membrane / initiate host damage . A6] receptor-mediated endocytosis A7] Fusion occurs between the membrane of the endosome and viral envelope . The capsid is released into the cytosol A8] Host Cellular enzymes hydrolyse / remove the capsid releasing viral genome / RNA
Mode of replication of genome	A9] Advantage of rapid replication of viral genome using host DNA dependent DNA polymerase during Lytic cycle D11] Disadvantage in lytic cycle as it causes the rapid destruction of host cells .	A10] Advantage of rapid replication of viral genome using host RNA dependent RNA polymerase. D11] Disadvantage in as it causes the rapid destruction of host cells.
Rate of production of viral copies	A12] Rapid release of viral copies all at the same time upon lysis of bacterial cell.	D13] Disadvantage in a relatively slower rate of release of viral copies via budding.
Number of modes of reproduction	A14] Lambda phage has the advantage of switching between two modes of reproduction that better allows a more sustained continuation of reproduction.	D15] Influenza has only a single mode of reproduction .
Effort on the part of virus	A16] Lambda through the lysogenic cycle is able to passively (through no active action on its part) reproduce copies of its genome.	D17] Influenza's reproductive cycle requires active intervention to reproduce copies of its genome.
Other less viable points	R1] Advantage of slow sustained replication of viral genome via binary fission of bacteria during lysogenic cycle . R2] integration of viral genome into bacterial genome via viral enzyme integrase .	No equal level of comparison.

(b) Suggest how phenotypic variation is brought about in viruses

[12]

ANS [L2/3] (Novel)

[12]

SC:	Mode	Mutation	Genotype Variation	Mechanism	phenotypic variation
OR:	1] Genetic Drift	2] Point Mutation	Base Pair Substitution	3] mistakes by DNA Poly I and III	4] change in single aa
			5] Addition	7] mistake by DNA Poly I and III	8] change in polypep
			6] Deletion		9] Frame shift
	10] Genetic Shift		11] Virus with >2 chromosomes		
			12] genetically diff strains infect same host		
				13] mistakes in spontaneous assembly	14] Phenotypically diff

- Genetic Drift** as a more common mode to produce subtly different genetic sequences that may produce phenotypic variation.
- Point mutation as a mechanism of Genetic drift- base pair substitution of DNA
- Brought about thru mistakes in replication of viral genome via DNA Poly I / III or RNA dependent RNA polymerases.
- Causes a change in a single amino acid that may result in variation in phenotype.
- Addition as the 2nd mechanism of Genetic drift
- Deletion as the 3rd Mechanism of genetic drift
- Both Addition and deletion are brought about by mistakes via DNA Poly I / III or RNA dependent RNA polymerases.
- Caused by the change in the reading frame as a result of addition or deletion
- Result in changes in the section of polypeptide downstream of mutation.
- Genetic Shift** as a more uncommon mode
- Occurring in viruses possessing more than 2 chromosomes
- Genetically different strains of viruses infect the same host cell
- Through mistakes in spontaneous assembly of viruses – attain phenotypically different viruses.
- evolutionary causes to changes in phenotype- via inheritance
- OWTTE.
- AVP

[Total: 25]

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